GOOGLE ADS SALES DATA ANALYSIS

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1. Dataset Used

For this project, I used the Google Ads Sales dataset from Kaggle. The dataset had 2,600 records and different columns related to advertisements, such as clicks, impressions, cost, leads, conversions, sales, date, location, device, and keyword.

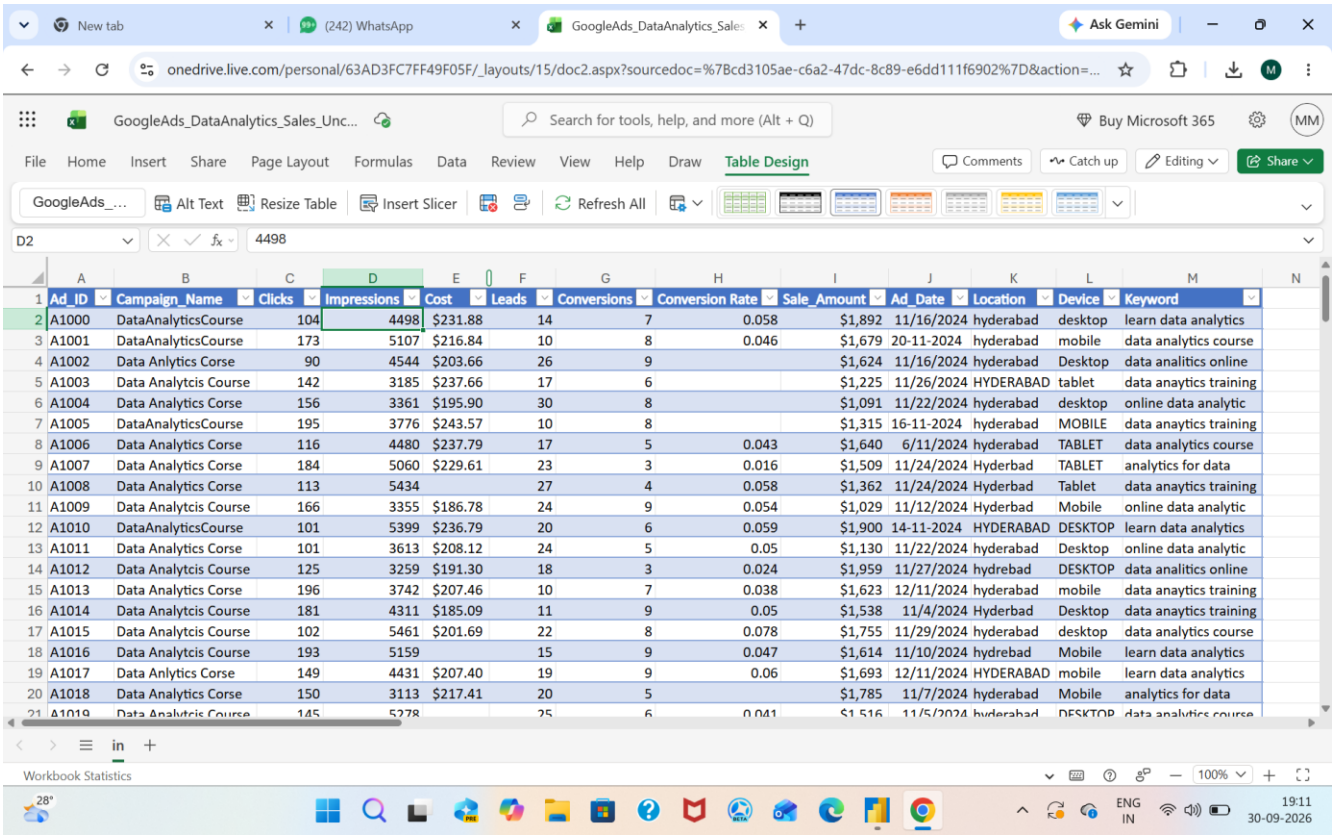
Dataset link: https://www.kaggle.com/datasets/nayakganesh007/google-ads-sales-dataset

2. Opened the Raw Dataset

First, I opened the downloaded dataset in Excel. I kept the original data separately in the Raw Data sheet so that I would not change the original dataset while doing the cleaning.

3. Converted the Data into an Excel Table

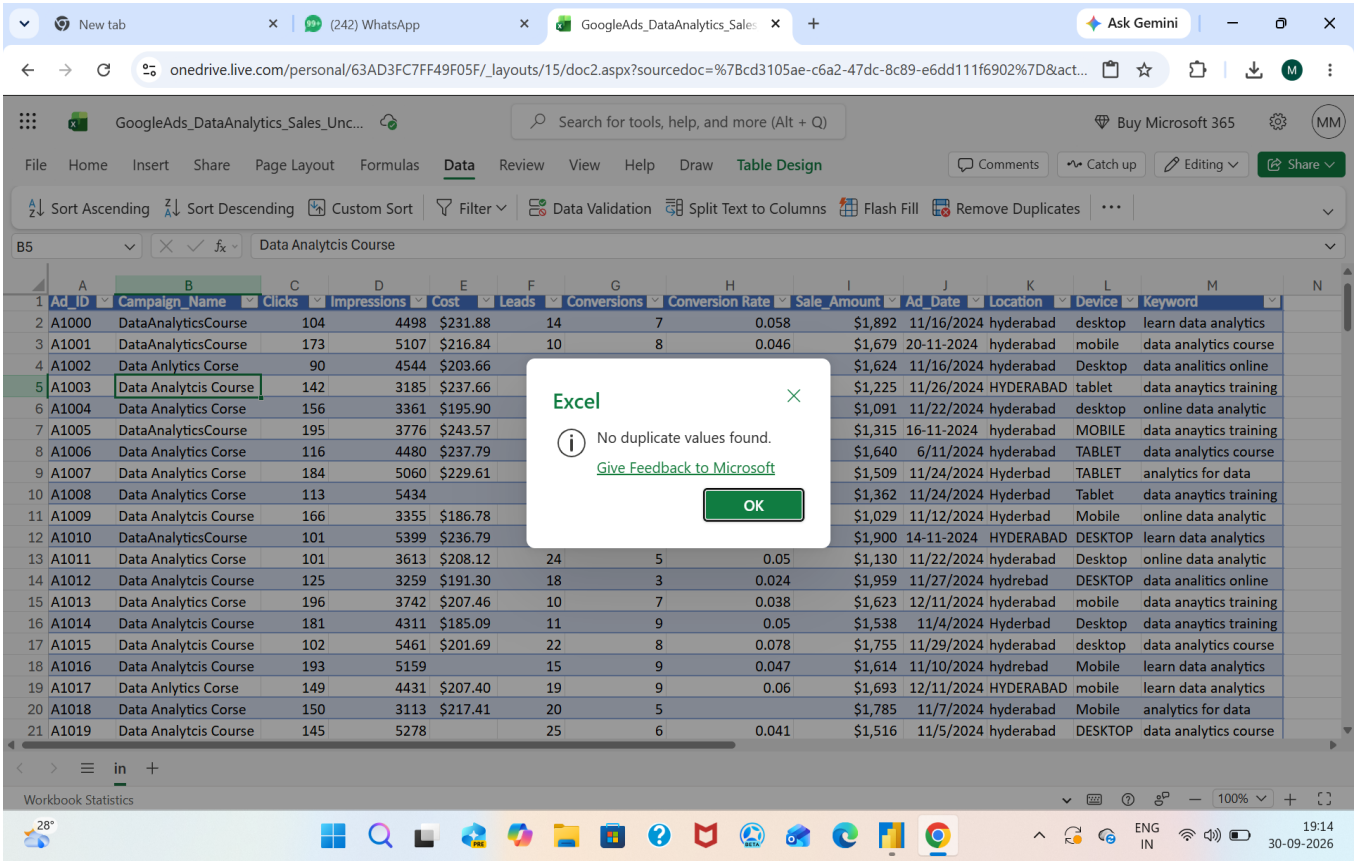
I selected the complete dataset and converted it into an Excel Table. This made it easier to filter the columns, work with the records and manage the data during the analysis.



Ad_ID	Campaign_Name	Clicks	Impressions	Cost	Leads	Conversions	Conversion Rate	Sale_Amount	Ad_Date	Location	Device	Keyword
A1000	DataAnalyticsCourse	104	4498	$231.88	14	7	0.058	$1,892	11/16/2024	hyderabad	desktop	learn data analytics
A1001	DataAnalyticsCourse	173	5107	$216.84	10	8	0.046	$1,679	20-11-2024	hyderabad	mobile	data analytics course
A1002	Data Anlytics Corse	90	4544	$203.66	26	9		$1,624	11/16/2024	hyderabad	Desktop	data analitics online
A1003	Data Analytcis Course	142	3185	$237.66	17	6		$1,225	11/26/2024	HYDERABAD	tablet	data anaytics training
A1004	Data Analytics Corse	156	3361	$195.90	30	8		$1,091	11/22/2024	hyderabad	desktop	online data analytic
A1005	DataAnalyticsCourse	195	3776	$243.57	10	8		$1,315	16-11-2024	hyderabad	MOBILE	data anaytics training
A1006	Data Analytics Corse	116	4480	$237.79	17	5	0.043	$1,640	6/11/2024	hyderabad	TABLET	data analytics course
A1007	Data Analytics Corse	184	5060	$229.61	23	3	0.016	$1,509	11/24/2024	Hyderbad	TABLET	analytics for data
A1008	Data Analytics Corse	113	5434		27	4	0.058	$1,362	11/24/2024	Hyderbad	Tablet	data anaytics training
A1009	Data Analytcis Course	166	3355	$186.78	24	9	0.054	$1,029	11/12/2024	Hyderbad	Mobile	online data analytic
A1010	DataAnalyticsCourse	101	5399	$236.79	20	6	0.059	$1,900	14-11-2024	HYDERABAD	DESKTOP	learn data analytics
A1011	Data Analytics Corse	101	3613	$208.12	24	5	0.05	$1,130	11/22/2024	hyderabad	Desktop	online data analytic
A1012	Data Analytcis Course	125	3259	$191.30	18	3	0.024	$1,959	11/27/2024	hydrebad	DESKTOP	data analitics online
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A1014	Data Analytcis Course	181	4311	$185.09	11	9	0.05	$1,538	11/4/2024	Hyderbad	Desktop	data anaytics training
A1015	Data Analytcis Course	102	5461	$201.69	22	8	0.078	$1,755	11/29/2024	hyderabad	desktop	data analytics course
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4. Checked for Duplicate Records

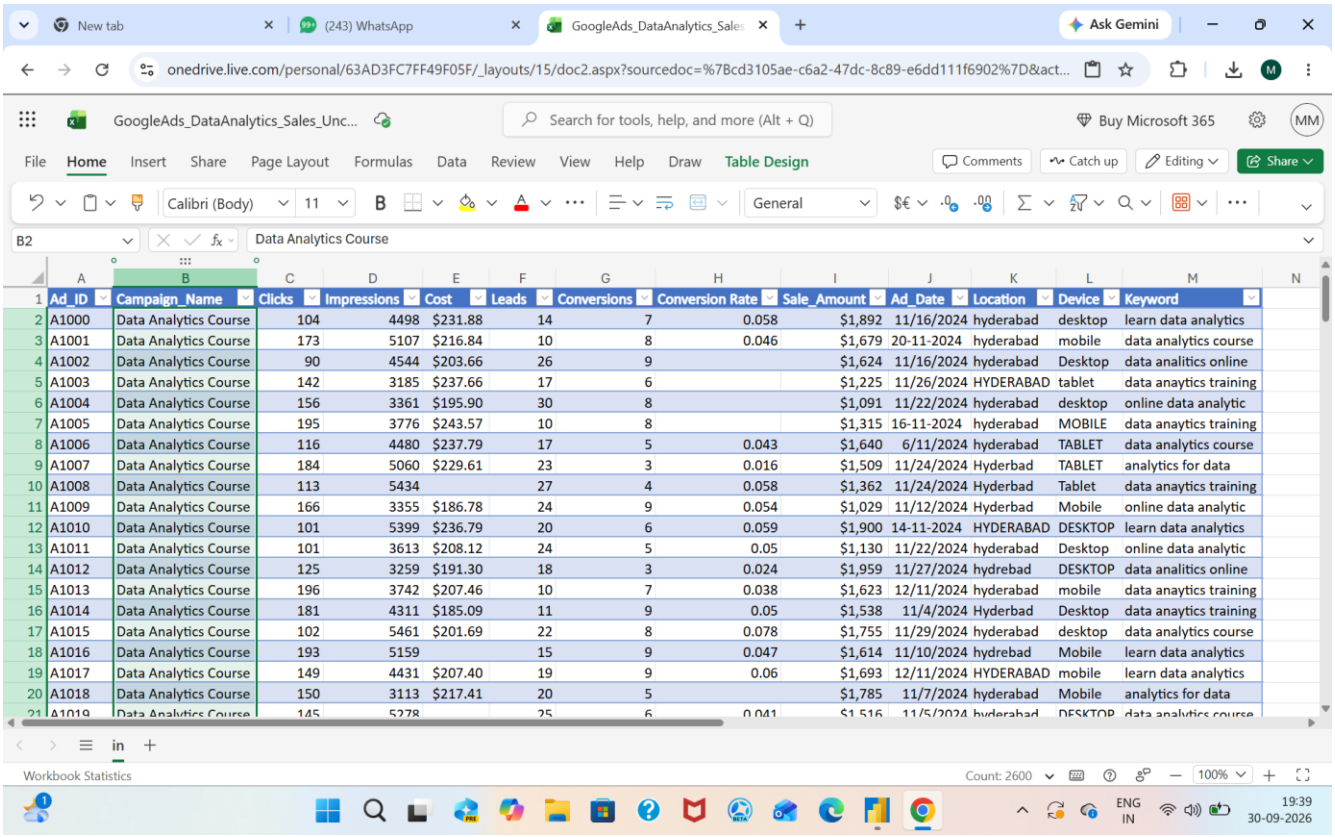
I checked the Ad_ID column for duplicate values. The purpose was to make sure that the same advertisement record was not repeated. No duplicate Ad_ID records were found.



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5. Cleaned Campaign Names

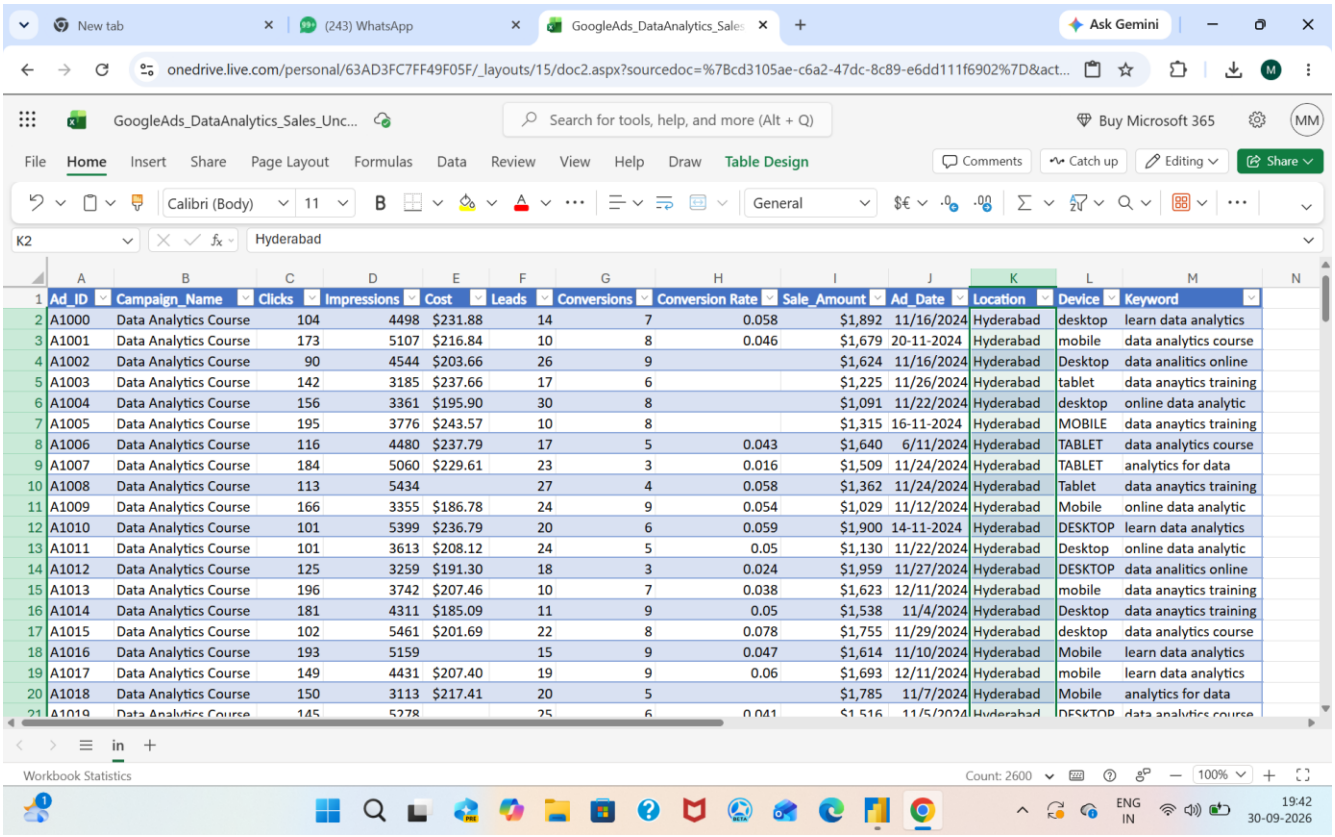
The Campaign_Name column had different spellings for the same campaign. For example, 'Data Analytcis Course', 'Data Analytics Corse', 'DataAnalyticsCourse' and 'Data Anlytics Corse' were present. I standardized these values to 'Data Analytics Course'.



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6. Cleaned Location Names

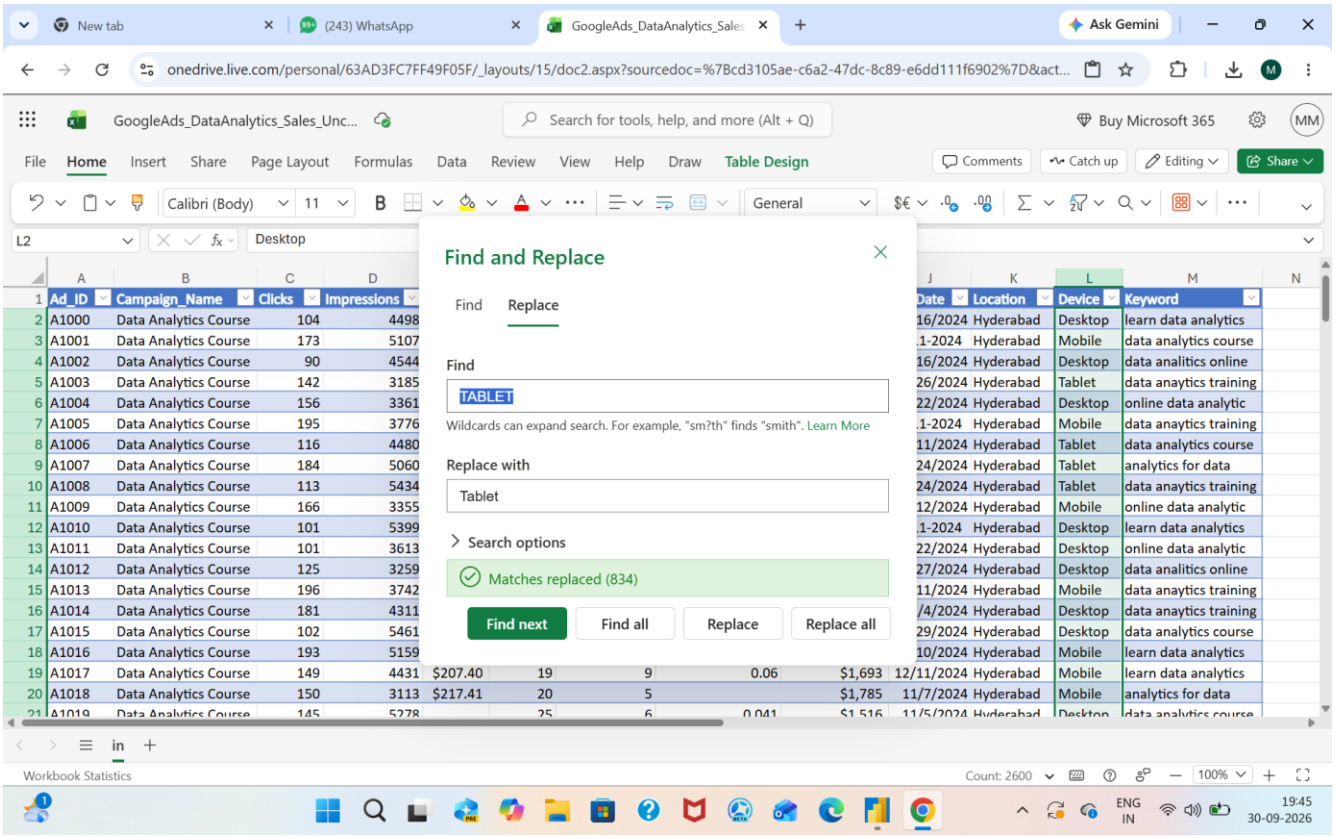
The Location column also had inconsistent spellings and capitalization. Values such as 'hyderabad', 'HYDERABAD', 'Hyderbad' and 'hydrebad' were standardized to 'Hyderabad'.



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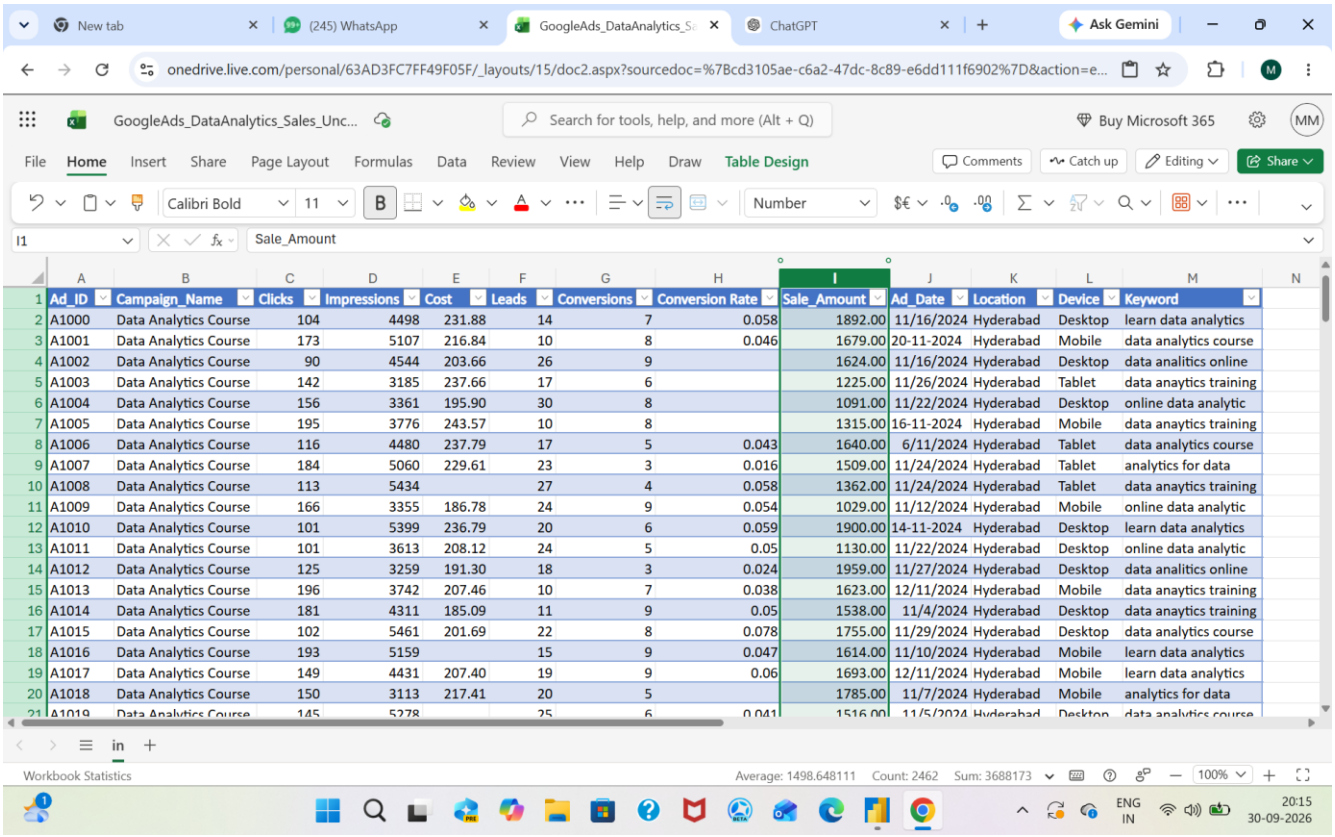
7. Standardized Device Names

The Device column contained the same device in different cases, such as Desktop, desktop, and DESKTOP. I standardized the values into three consistent categories: Desktop, Mobile, and Tablet.



8. Cleaned Cost and Sale Amount

Cost and Sale Amount contained values with currency symbols or text formatting. I cleaned and formatted these columns as numbers so that they could be used correctly in calculations.

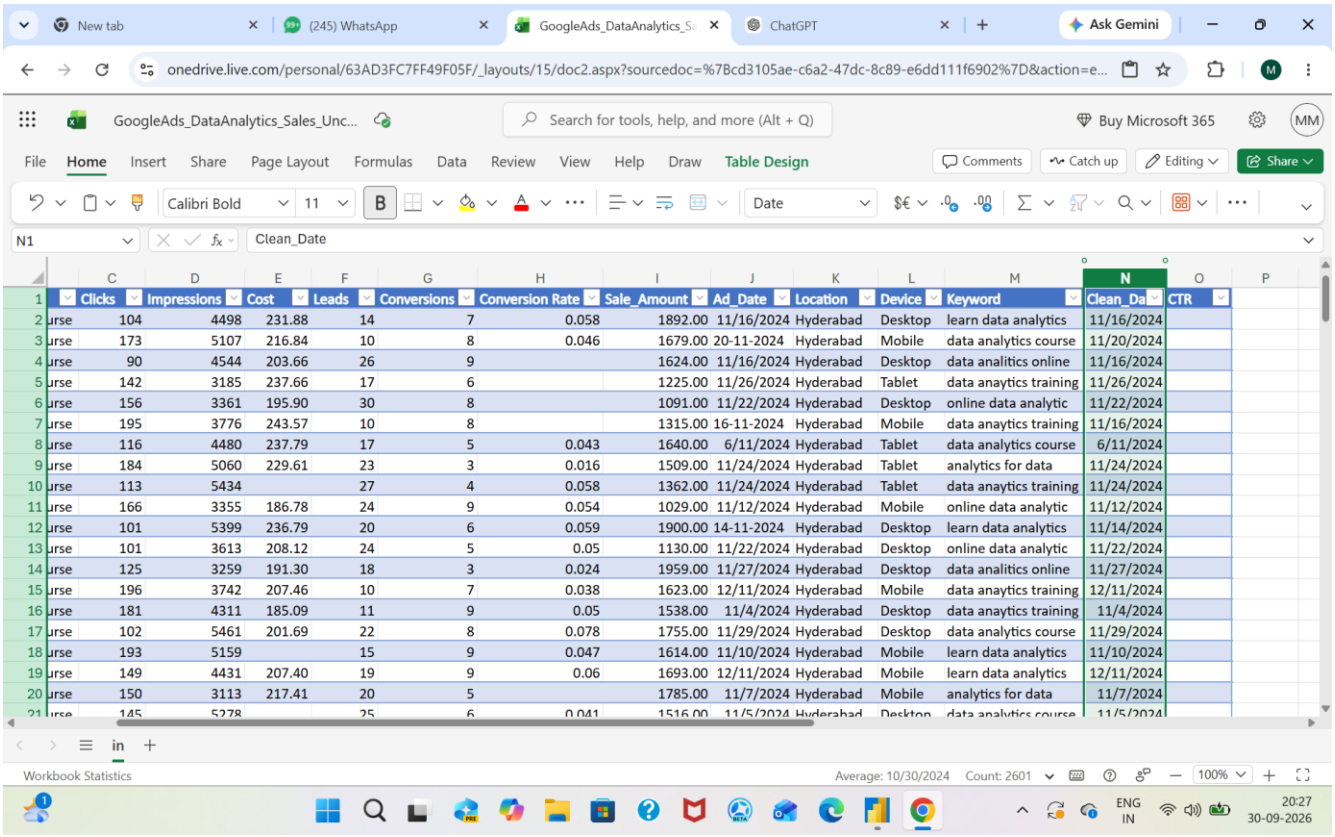


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9. Cleaned the Date Column

The Ad_Date column contained different date formats, for example dates written with '-' and '/'. I created a new Clean_Date column and used a formula to convert the different formats into one consistent date format.

The final date was displayed in the format dd-mmm-yyyy, for example 16-Nov-2024.

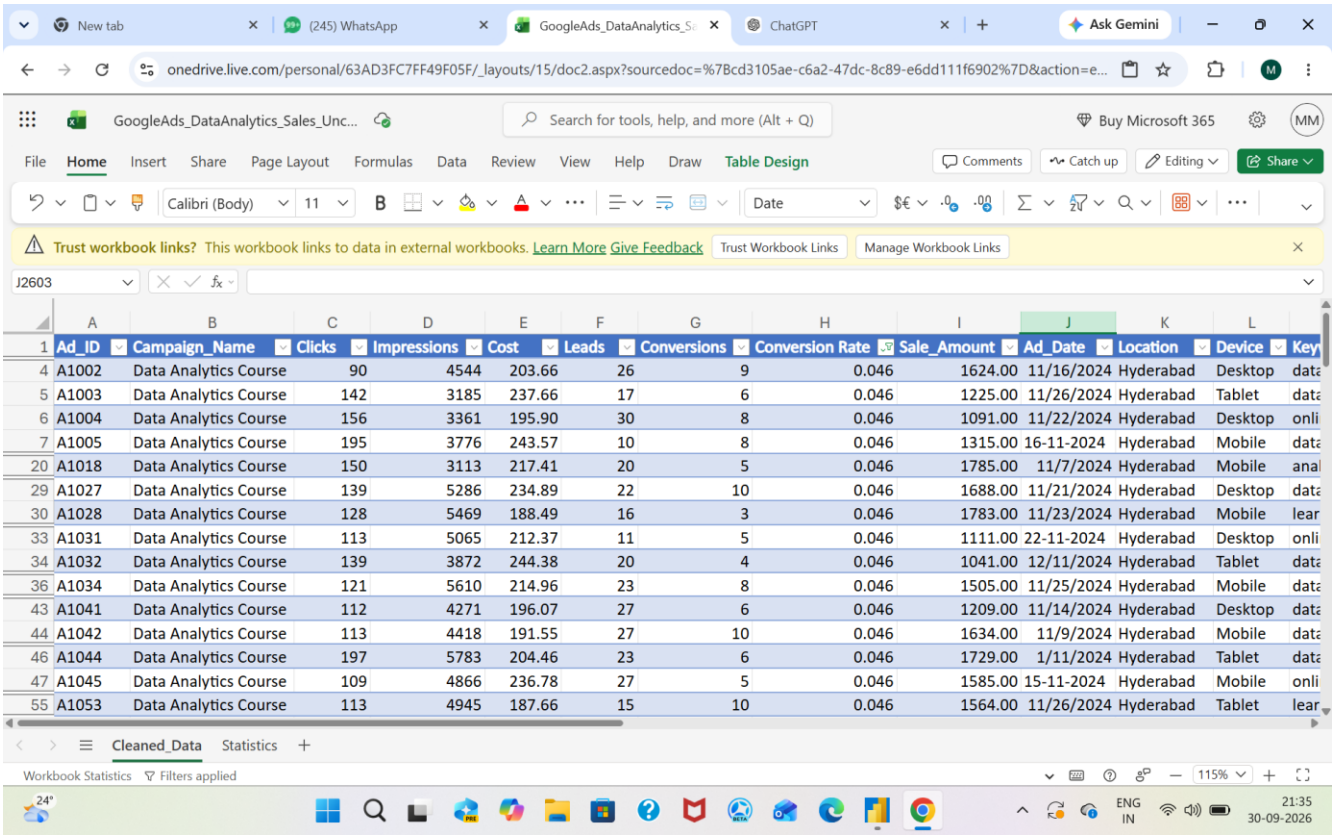


	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1	Clicks	Impressions	Cost	Leads	Conversions	Conversion Rate	Sale_Amount	Ad_Date	Location	Device	Keyword	Clean_Da	CTR	
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21	145	5278		25	6	0.041	1516.00	11/5/2024	Hyderabad	Desktop	data analytics course	11/5/2024		

10. Handled Missing Values

I checked the dataset for blank values. Missing values were present in numerical columns such as Clicks, Impressions, Cost, Leads, Conversions, Sale_Amount and Conversion Rate.

Instead of deleting the records, I used the median value of each respective column to fill out the blanks. This helped me retain all 2,600 records.



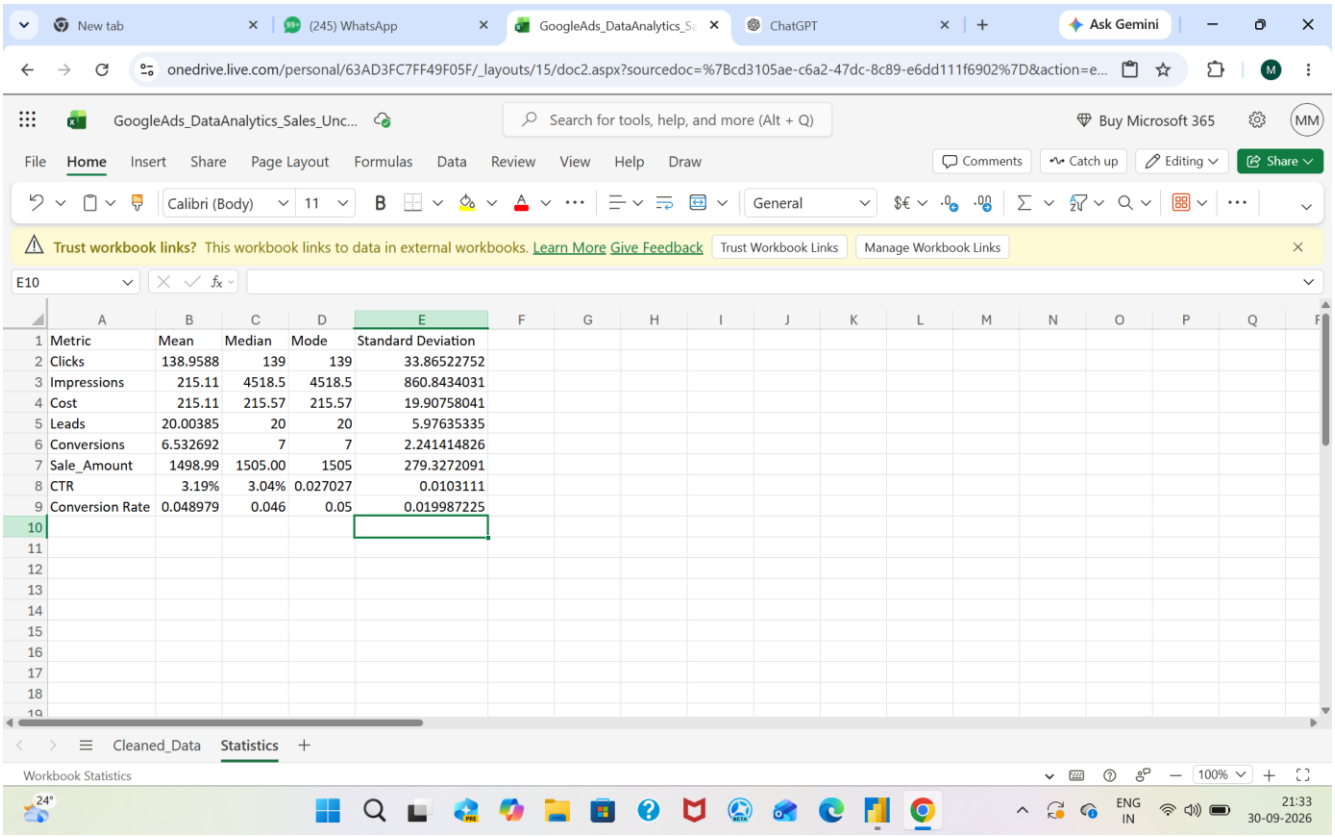
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A1004	Data Analytics Course	156	3361	195.90	30	8	0.046	1091.00	11/22/2024	Hyderabad	Desktop	onli
A1005	Data Analytics Course	195	3776	243.57	10	8	0.046	1315.00	16-11-2024	Hyderabad	Mobile	data
A1018	Data Analytics Course	150	3113	217.41	20	5	0.046	1785.00	11/7/2024	Hyderabad	Mobile	anal
A1027	Data Analytics Course	139	5286	234.89	22	10	0.046	1688.00	11/21/2024	Hyderabad	Desktop	data
A1028	Data Analytics Course	128	5469	188.49	16	3	0.046	1783.00	11/23/2024	Hyderabad	Mobile	lear
A1031	Data Analytics Course	113	5065	212.37	11	5	0.046	1111.00	22-11-2024	Hyderabad	Desktop	onli
A1032	Data Analytics Course	139	3872	244.38	20	4	0.046	1041.00	12/11/2024	Hyderabad	Tablet	data
A1034	Data Analytics Course	121	5610	214.96	23	8	0.046	1505.00	11/25/2024	Hyderabad	Mobile	data
A1041	Data Analytics Course	112	4271	196.07	27	6	0.046	1209.00	11/14/2024	Hyderabad	Desktop	data
A1042	Data Analytics Course	113	4418	191.55	27	10	0.046	1634.00	11/9/2024	Hyderabad	Mobile	data
A1044	Data Analytics Course	197	5783	204.46	23	6	0.046	1729.00	1/11/2024	Hyderabad	Tablet	data
A1045	Data Analytics Course	109	4866	236.78	27	5	0.046	1585.00	15-11-2024	Hyderabad	Mobile	onli
A1053	Data Analytics Course	113	4945	187.66	15	10	0.046	1564.00	11/26/2024	Hyderabad	Tablet	lear

11. Created Calculated Columns

After cleaning the data, I created a few calculated columns to analyse advertising performance.

Column
CTR
CPC
CPL
CPCONV
ROAS

I used IFERROR in these calculations so that the formula would return 0 instead of showing an error when the denominator was zero.



Metric	Mean	Median	Mode	Standard Deviation
Clicks	138.9588	139	139	33.86522752
Impressions	215.11	4518.5	4518.5	860.8434031
Cost	215.11	215.57	215.57	19.90758041
Leads	20.00385	20	20	5.97635335
Conversions	6.532692	7	7	2.241414826
Sale_Amount	1498.99	1505.00	1505	279.3272091
CTR	3.19%	3.04%	0.027027	0.0103111
Conversion Rate	0.048979	0.046	0.05	0.019987225

12. Created the Statistics Sheet

I created a separate Statistics sheet to calculate descriptive statistics for the main numerical fields. I calculated Mean, Median, Mode, and Standard Deviation for Clicks, Impressions, Cost, Leads, Conversions, Sale_Amount, CTR, and Conversion Rate.

I used AVERAGE, MEDIAN, MODE.SNGL and STDEV.S functions. The formulas were applied only to rows 2 to 2601 because the dataset contains 2,600 actual records.

13. Used SUM Function

I used the SUM function to find the total Clicks, total Cost and total Sale Amount in the dataset.

- Total Clicks = 361,293
- Total Cost = 559,282.12
- Total Sale Amount = 3,897,368

14. Used COUNTIF Function

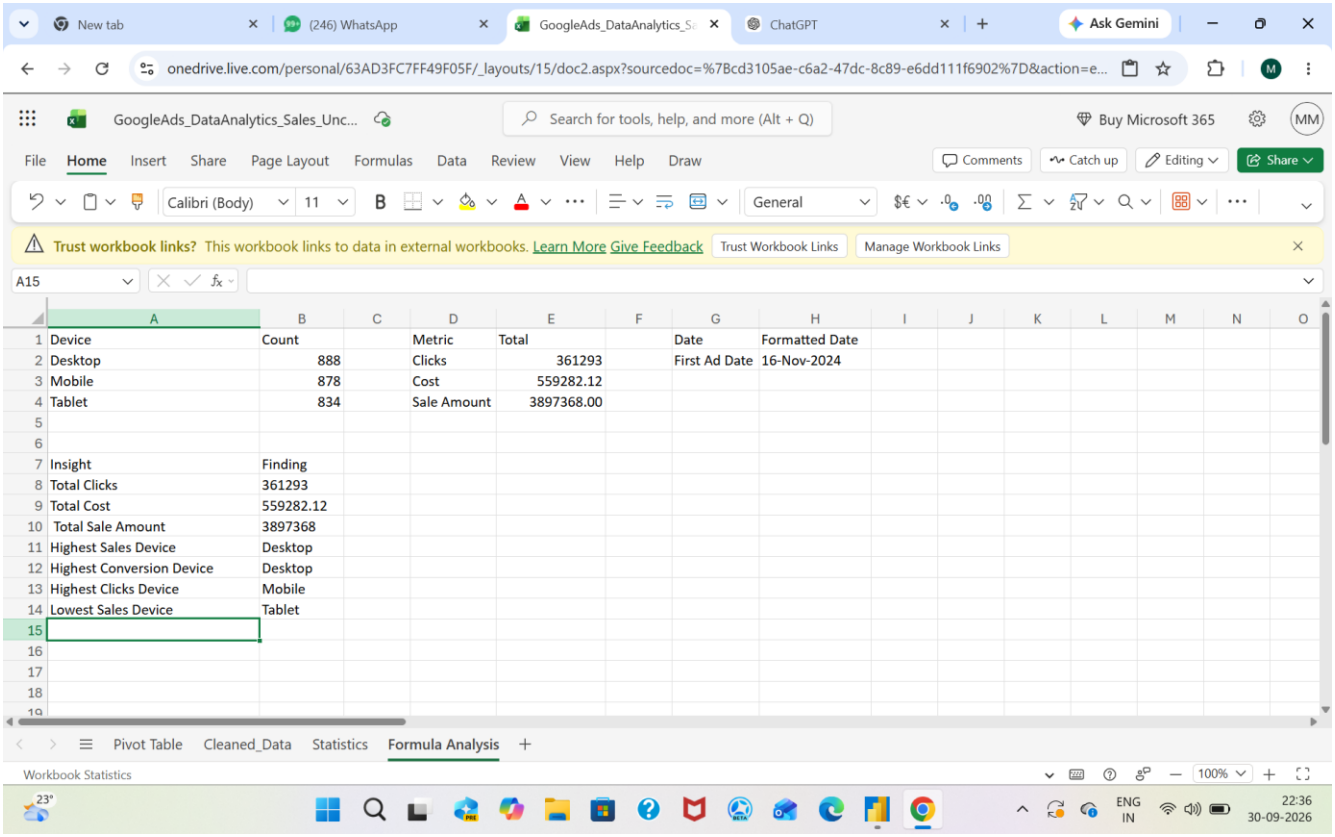
I used COUNTIF to count how many records belonged to each device category. The data range was kept fixed while copying the formula.

- Desktop = 888
- Mobile = 878

- Tablet = 834

15. Used TEXT Function

I used the TEXT function to display the cleaned date in a readable format. For example, the date was displayed as 16-Nov-2024 instead of the original mixed formats.

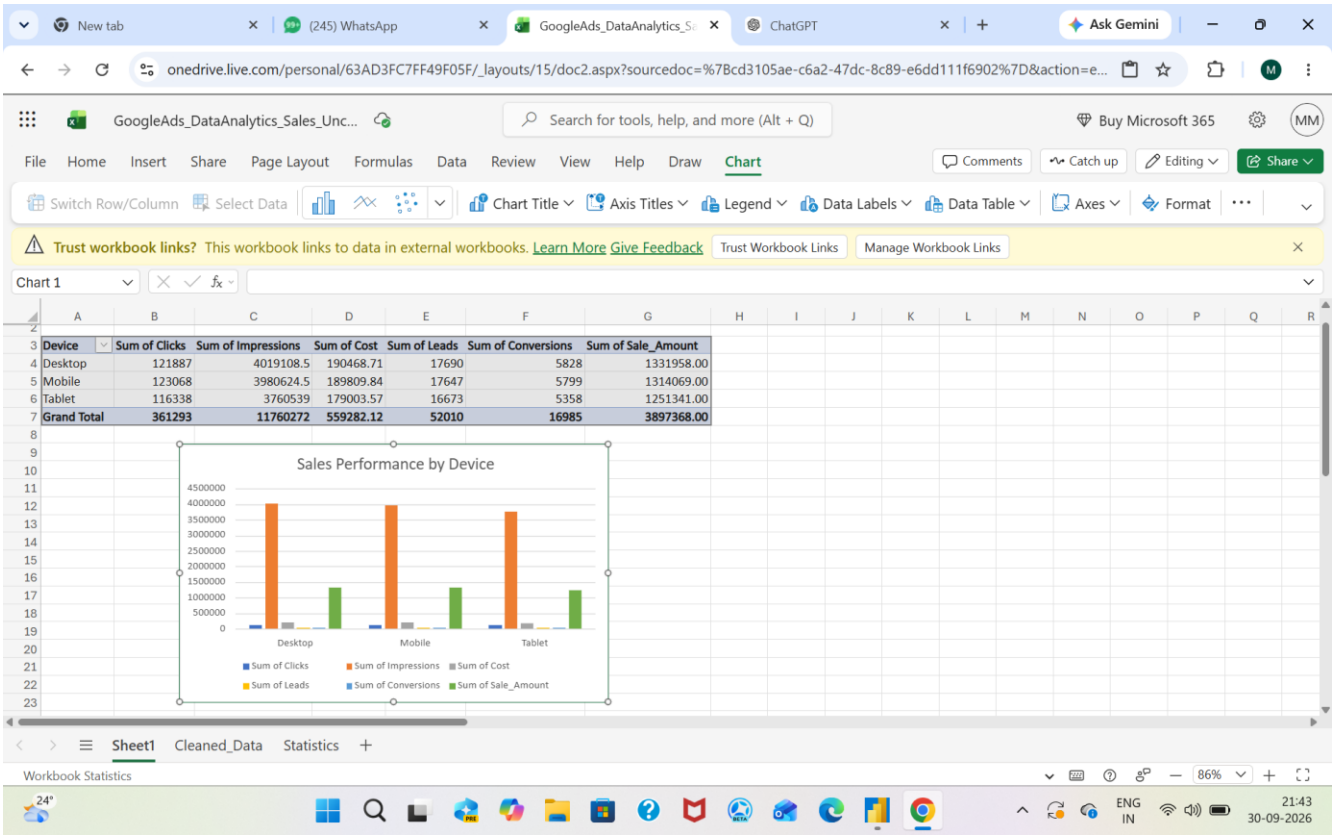


	A	B	C	D	E	F	G	H
1	Device	Count		Metric	Total		Date	Formatted Date
2	Desktop	888		Clicks	361293		First Ad Date	16-Nov-2024
3	Mobile	878		Cost	559282.12			
4	Tablet	834		Sale Amount	3897368.00			
5								
6								
7	Insight	Finding						
8	Total Clicks	361293						
9	Total Cost	559282.12						
10	Total Sale Amount	3897368						
11	Highest Sales Device	Desktop						
12	Highest Conversion Device	Desktop						
13	Highest Clicks Device	Mobile						
14	Lowest Sales Device	Tablet						
15								
16								
17								
18								

16. Created the PivotTable

After completing the cleaning and formula work, I created a PivotTable from the cleaned dataset. I used Device as the row field and added Clicks, Impressions, Cost, Leads, Conversions and Sale_Amount as value fields.

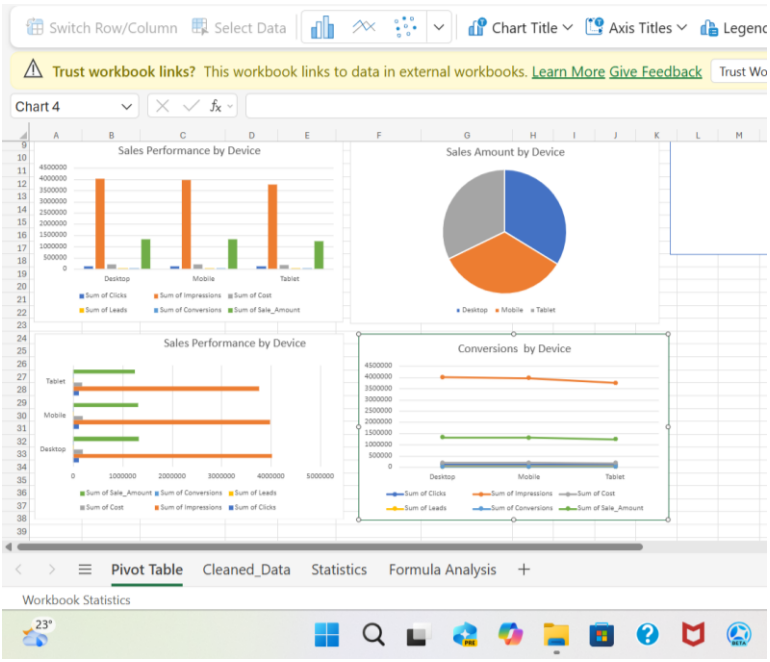
This helped me compare the advertising performance of Desktop, Mobile, and Tablet.



17. Created the Charts

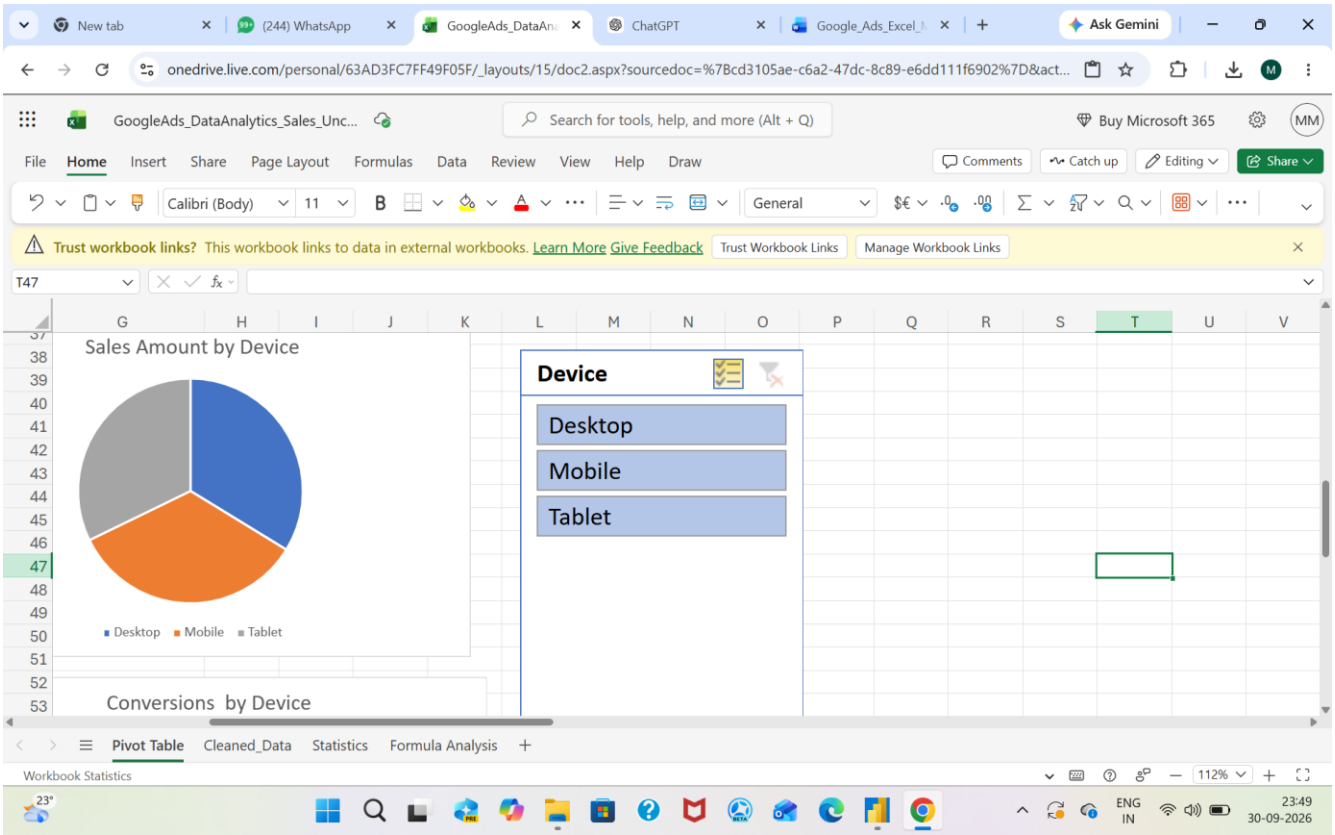
I created three different chart types from PivotTable so that the results could be understood visually.

- Clustered Column Chart – Sales Performance by Device
- Pie Chart – Sales Amount by Device
- Line Chart – Conversions by Device
- Bar Chart – Sales Performance by Device



18. Added a Device Slicer

I added a Device slicer to the dashboard. It allows the user to select Desktop, Mobile, or Tablet and view the related information interactively.



19. Final Excel Dashboard

Finally, I arranged the PivotTable, charts and slicer on Sheet 1. The purpose of the dashboard is to give a simple visual comparison of advertising performance by device.

20. Main Findings from the Excel Analysis

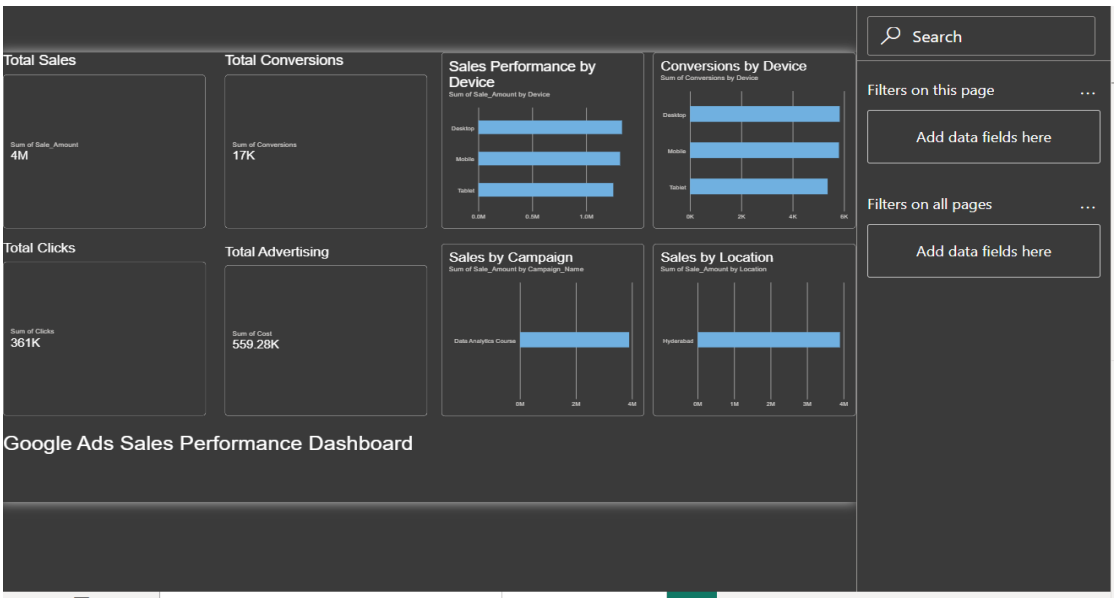
From the analysis, Desktop recorded the highest sales amount and conversions, while Mobile recorded the highest number of clicks. The tablet recorded the lowest sales amount among the three devices. These comparisons will be used as a starting point for the Power BI dashboard.

21. Next Step – Power BI After completing the Excel cleaning and analysis, the cleaned dataset will be used in Power BI. The next stage is to create an interactive Power BI dashboard using suitable charts, cards, and slicers.

> Power BI Report The cleaned Excel data was imported into Power BI. The report was divided into two pages to keep the dashboard readable and interactive.

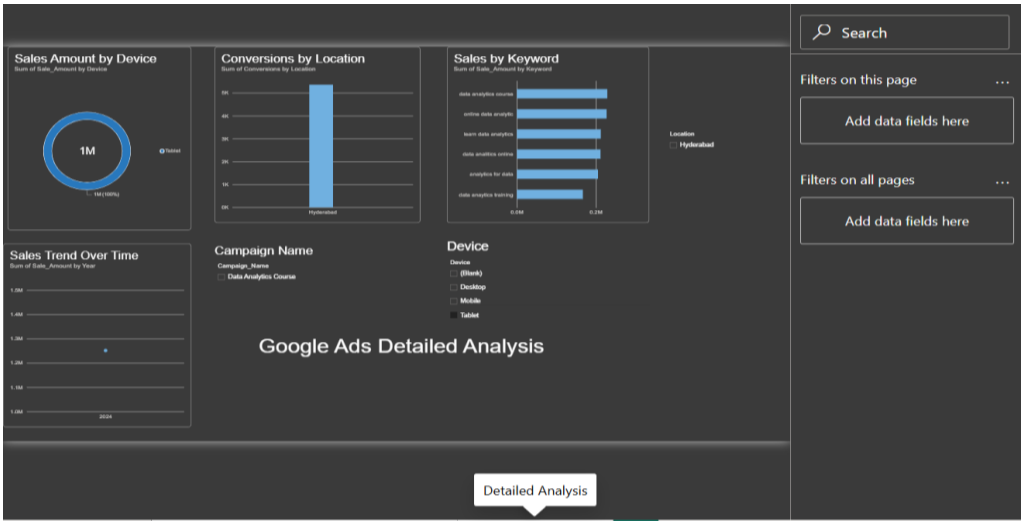
Page 1 – Google Ads Sales Performance Dashboard

• Total Sales card • Total Conversions card • Total Clicks card • Total Advertising Cost card • Sales Performance by Device • Conversions by Device • Sales by Campaign • Sales by Location.



Page 2 – Detailed Analysis

• Sales Amount by Device – donut chart • Conversions by Location • Sales by Keyword • Sales Trend Over Time • Device slicer • Location slicer • Campaign Name slicer



Main Findings

• Desktop generated the highest sales. • Desktop generated the highest conversions. • Mobile received the highest clicks. • Tablet had the lowest sales and conversions among the three devices. • Advertising performance can be compared across devices, campaigns and locations. The device with the highest clicks is not automatically the best-performing device; sales, conversions and advertising costs should also be considered.

22. Conclusion

The project demonstrated the complete process of preparing advertising data for analysis. Excel was used for cleaning, standardization, missing-value handling, calculations, statistics, and initial analysis. Power BI was then used to create an interactive report with cards, charts, and slicers.